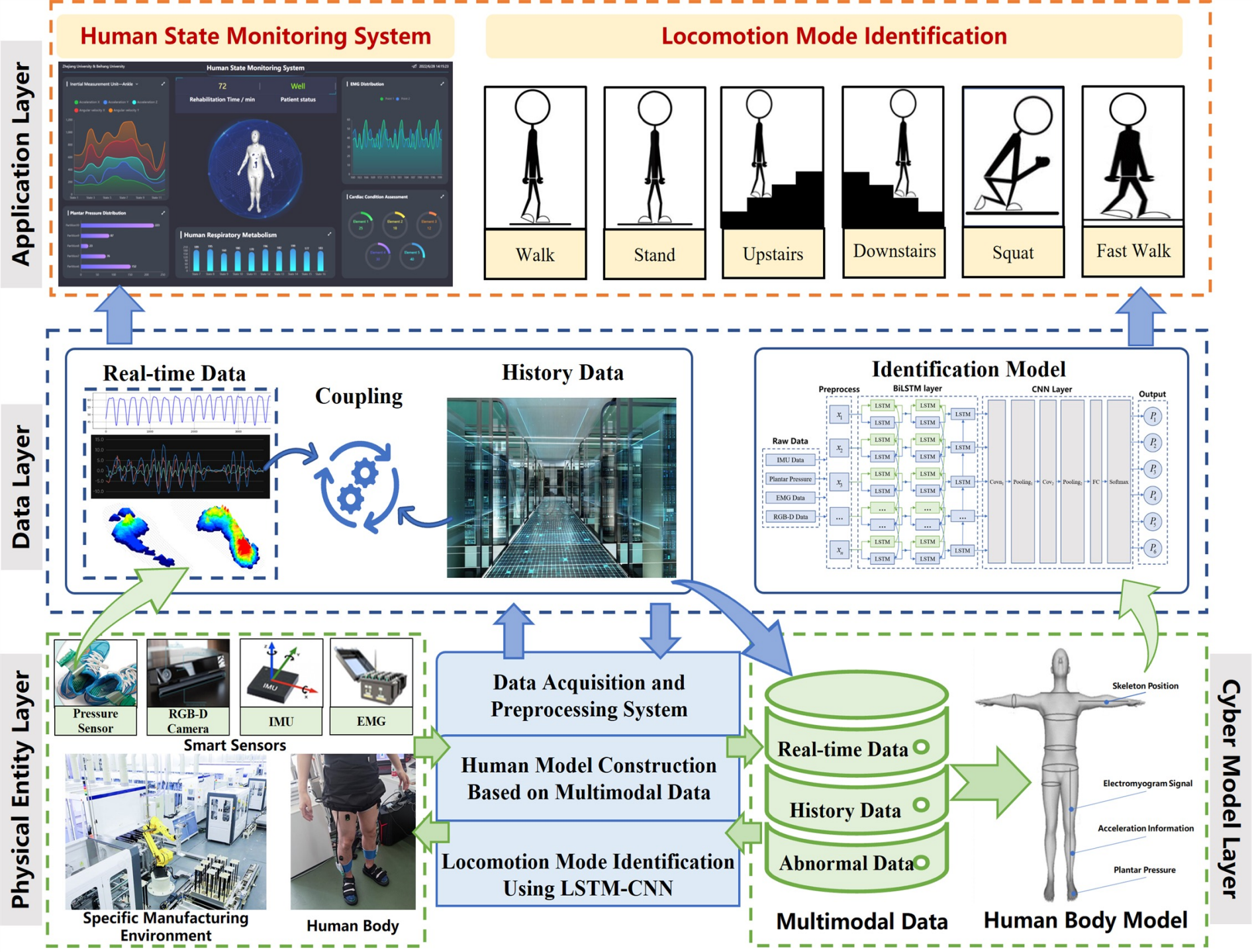
Application Layer
Human State Monitoring System
Locomotion Mode Identification
Walk
Stand
Upstairs
Downstairs
Squat
Fast Walk
Data Layer
Real-time Data
Coupling
History Data
Identification Model
Preprocess
BiLSTM layer
CNN Layer
Output
Raw Data
IMU Data
Plantar Pressure
EMG Data
RGB-D Data
LSTM
Softmax
FC
Physical Entity Layer
Pressure Sensor
RGB-D Camera
IMU
EMG
Smart Sensors
Specific Manufacturing Environment
Human Body
Data Acquisition and Preprocessing System
Human Model Construction Based on Multimodal Data
Locomotion Mode Identification Using LSTM-CNN
Real-time Data
History Data
Abnormal Data
Multimodal Data
Skeleton Position
Electromyogram Signal
Acceleration Information
Plantar Pressure
Human Body Model
Cyber Model Layer